

EXPERIMENT-12

Student Name: Gurkirat Singh Bhangoo
Branch: CSE-AIML
Semester: 5
Subject Name: ADBMS

UID: 24BAI70370
Section: 24AIT_KRG-1_B
Date of Performance: 21/9/26

Q1. DESIGN A TRIGGER WHICH:

**--1. WHENEVER THERE IS A INSERTION ON STUDENT TABLE THEN, THE CURRENTLY
INSERTED OR DELETED**

--ROW SHOULD BE PRINTED AS IT AS ON THE OUTPUT CONSOLE WINDOW.

```
CREATE OR REPLACE FUNCTION fn_student_audit() RETURNS  
TRIGGER  
LANGUAGE plpgsql AS  
$$  
BEGIN  
    IF TG_OP = 'INSERT' THEN  
        RAISE NOTICE 'Inserted Row -> ID: %, Name: %, Age: %, Class: %', NEW.id,  
            NEW.name, NEW.age, NEW.class;  
        RETURN NEW;  
  
    ELSIF  
        TG_OP = 'DELETE' THEN  
        RAISE NOTICE 'Deleted Row -> ID: %, Name: %, Age: %, Class: %', OLD.id,  
            OLD.name, OLD.age, OLD.class;  
        RETURN OLD;  
    END IF;  
  
    RETURN NULL; END;  
$$
```

Q2. DESIGN A POSTGRESQL TRIGGERS THAT:

Whenever a new employee is inserted in `tbl_employee`, a record should be added to `tbl_employee_audit` like:

"Employee name <emp_name> has been added at <current_time>"

Whenever an employee is deleted from `tbl_employee`, a record should be added to `tbl_employee_audit` like:

"Employee name <emp_name> has been deleted at <current_time>" The

solution must use PostgreSQL triggers.

```
CREATE OR REPLACE FUNCTION audit_employee_changes()
RETURNS TRIGGER
LANGUAGE plpgsql AS
$$
```

```
BEGIN
```

```
IF TG_OP = 'INSERT' THEN
```

```
INSERT INTO tbl_employee_audit(message)
```

```
VALUES ('Employee name ' || NEW.emp_name || ' has been added at ' || NOW());
```

```
RETURN NEW;
```

```
ELSIF TG_OP = 'DELETE' THEN
```

```
INSERT INTO tbl_employee_audit(message)
```

```
VALUES ('Employee name ' || OLD.emp_name || ' has been deleted at ' || NOW());
```

```
RETURN OLD;
```

```
END IF;
```

```
RETURN NULL; END;
```

```
$$
```

```
CREATE TRIGGER trg_employee_audit AFTER
```

```
INSERT OR DELETE
```

```
ON
```

```
tbl_employee FOR
```

```
EACH ROW
```

```
EXECUTE FUNCTION audit_employee_changes();
```

```
--TESTING THE TRIGGER
```

```
-- Insert an employee
```

```
INSERT INTO tbl_employee(emp_name, emp_salary) VALUES ('Aman', 50000);
```

```
-- Delete an employee
```

```
DELETE FROM tbl_employee WHERE emp_name = 'Aman';
```

```
-- Check audit log
```

```
SELECT * FROM tbl_employee_audit;
```